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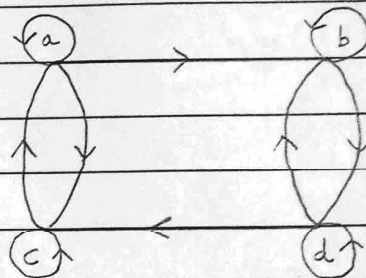
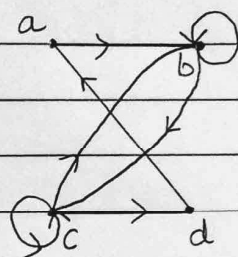
course program :- BTech CSE SY.

Subject :- Discrete Mathematics

Dm assignment 4.

Q.1) Find the adjacency matrix of the given directed multigraph with respect to the vertices listed in alphabetic order.

Sol:-



Sol:-

Vertices (in alphabetic order): a, b, c, d

from the graph (left diagram):

- $a \rightarrow b, a \rightarrow d$
- $b \rightarrow a, b \rightarrow c, b \rightarrow d$
- $c \rightarrow c$ (Self Loop), $c \rightarrow b, c \rightarrow d$

Now count the directed edge between every pair.

from \ to	a	b	c	d
a	0	1	0	1
b	1	0	1	1
c	0	1	1	1
d	0	1	1	0

Adjacency Matrix (A):

$$A = \begin{bmatrix} 6 & 1 & 6 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Q.2) Solve using Dijkstra's algorithm for following Undirected, Weighted Graph and find Lowest cost (ie. Shortest route) Path from A to F.

Sol:-

$$A-B=2, A-C=1, B-C=1$$

$$B-D=7, B-G=2, C-D=5$$

$$C-E=3, C-F=10, D-E=1$$

$$D-F=2, D-G=12, E-F=2$$

We will apply Dijkstra's Algorithm starting from A.

Step Vertex picked

start A

→ B

→ C

from C → D(6), E(4), F(11)

from B → G(4), D(9), C(1)

Smallest next = E(4)

from E → D(5), F(6)

Next = D(5)

from D $\rightarrow$ F(7), G(17)

Next = F(6)

Final Shortest distance $A \rightarrow F = 6$.

Shortest Path = $A \rightarrow C \rightarrow E \rightarrow F$

Q.3) Determine whether the given pair of graphs is isomorphic. Exhibit an isomorphism or provide a rigorous (detail) argument that, if not exists.

observations of both graphs

- Both have 6 vertices
- Each vertex has degree 3.
- Connectivity pattern and edges are identical (both appear as a regular 6-vertex graph with triangular inner connections).

Isomorphism mapping examples:

$u_1 \leftrightarrow v_1, u_2 \leftrightarrow v_2, u_3 \leftrightarrow v_3, u_4 \leftrightarrow v_4, u_5 \leftrightarrow v_5, u_6 \leftrightarrow v_6$

Hence, the graphs are isomorphic.

Q.4) Draw the graph whose incidence matrix is given below:-
Solⁿ:-

Each edge connects the two vertices having 1 in the same column:-

- $E_1 \rightarrow V_1$ - only V_1 has 1 $\rightarrow$ Self loop of V_1
- $E_2 \rightarrow V_1 - V_2$
- $E_3 \rightarrow V_1 - V_2 - V_3 \rightarrow$ invalid for simple graph
 $\rightarrow$ multigraph edge (maybe shared multiedge between $V_1 - V_2$)
- $E_4 \rightarrow V_2 - V_4$
- $E_5 \rightarrow V_3 - V_4$
- $E_6 \rightarrow V_2 - V_3 - V_5$ (implies edges among them likely two edges connecting $V_2 - V_3$ & $V_3 - V_5$).
- $E_7 \rightarrow V_3 - V_4$
- $E_8 \rightarrow V_1 - V_5$

* Graph features:-

- 5 vertices ($V_1 - V_5$)
- Mix of simple, multiedges, & a self loop of V_1 .

Q.11) Solve the following graph using Travelling Salesman problem.

Draw and calculate the shortest path.

Solⁿ:-

$$A - B = 16$$

$$B - C = 21$$

$$C - D = 12$$

$$D - E = 15$$

$$E - F = 16$$

$$F - A = 54$$

- diagonal $B-D = 9$
- diagonal $C-F = 7$.

(if there are other edges with different weights that I missed tell me i'll recompute)

Shortest tour found:

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow A$

Total cost:-

$$A-B(16) + B-C(21) + C-D(12) + D-E(15) + E-F(16) \\ + F-A(34) \Rightarrow \underline{\underline{114}}$$